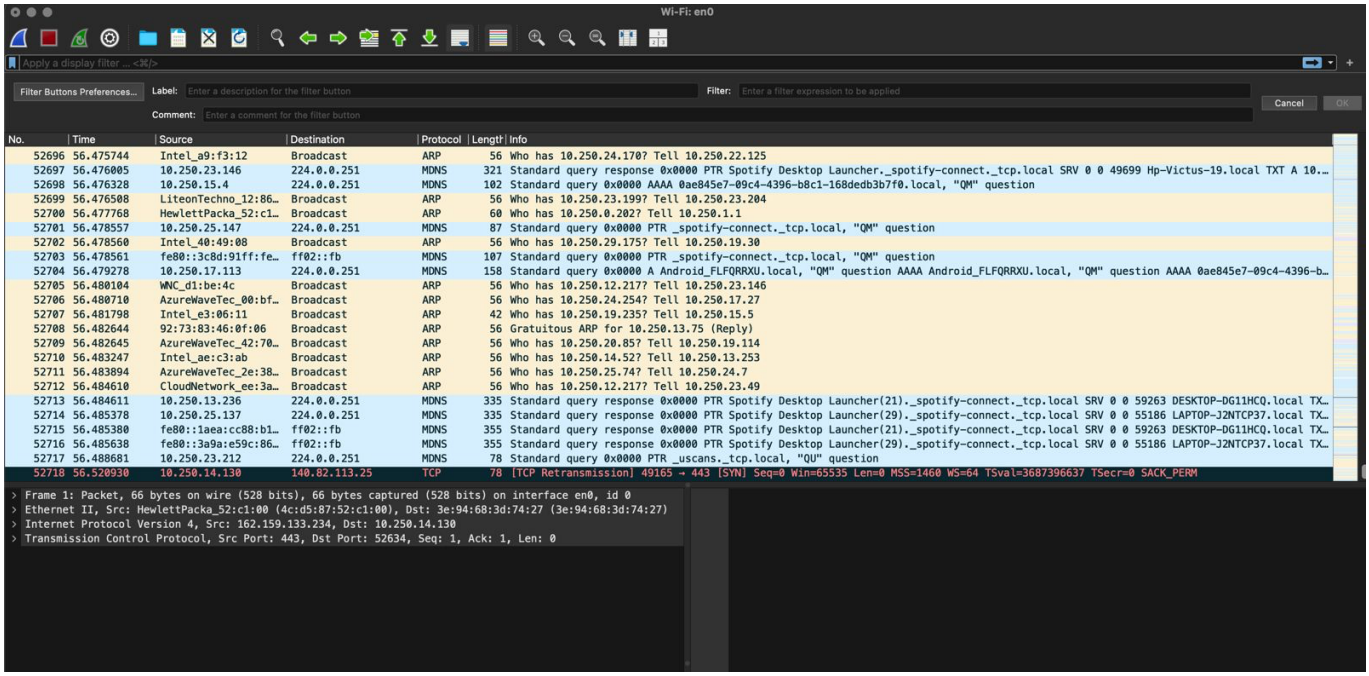
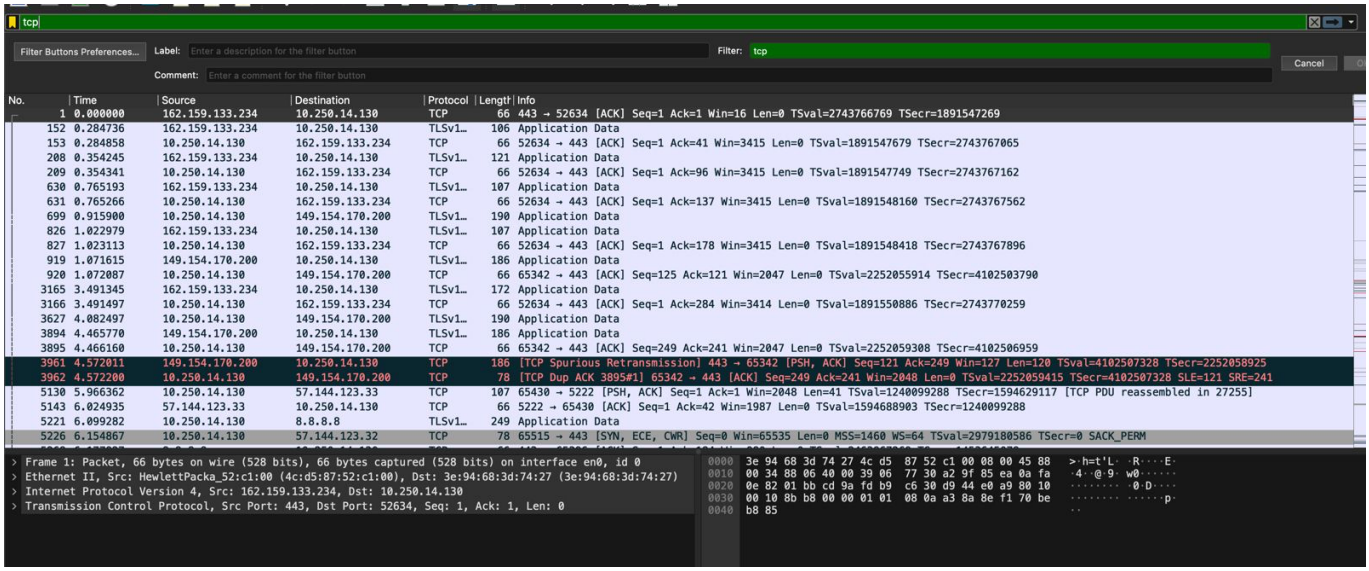
Skill 1 – Encapsulation Builder

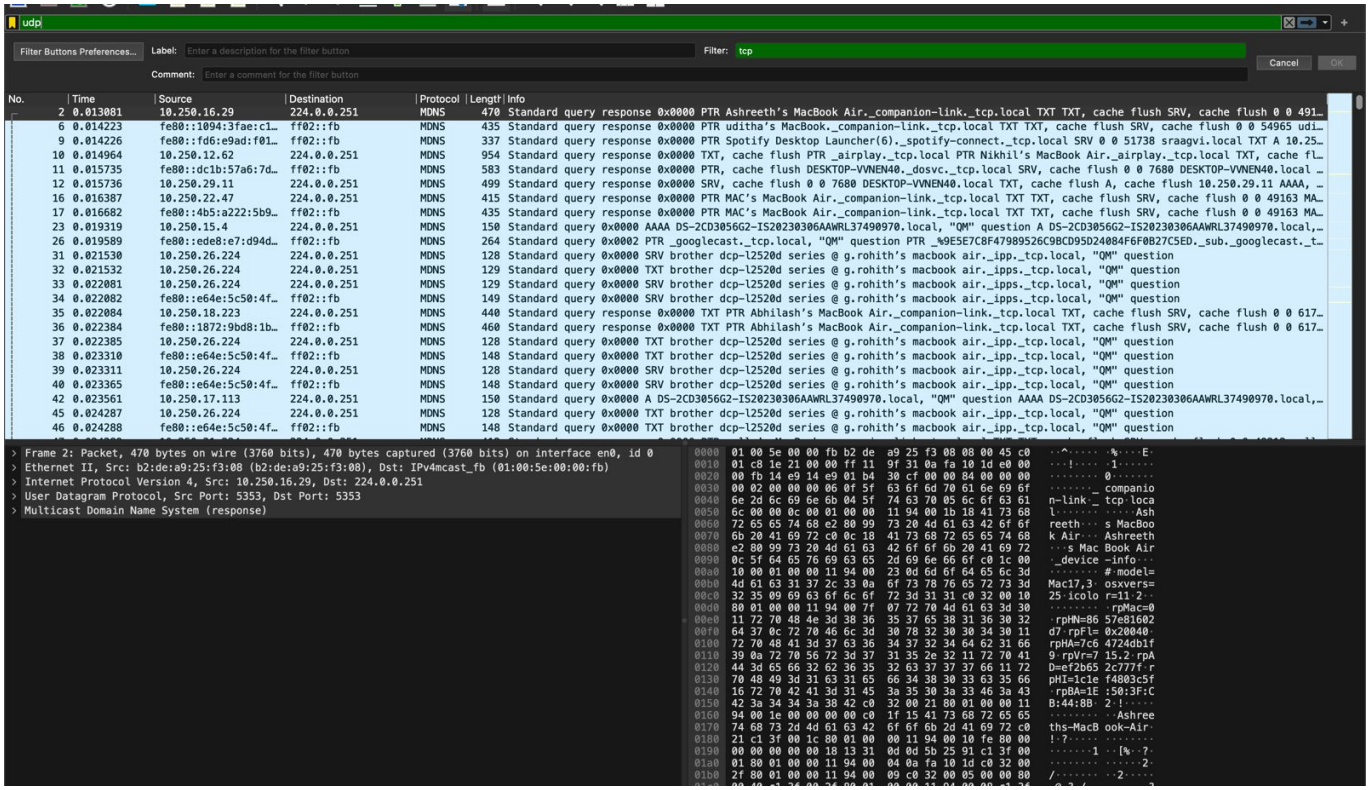
Step 1 & 2: Open Wireshark and start capturing on the Wi-Fi interface.

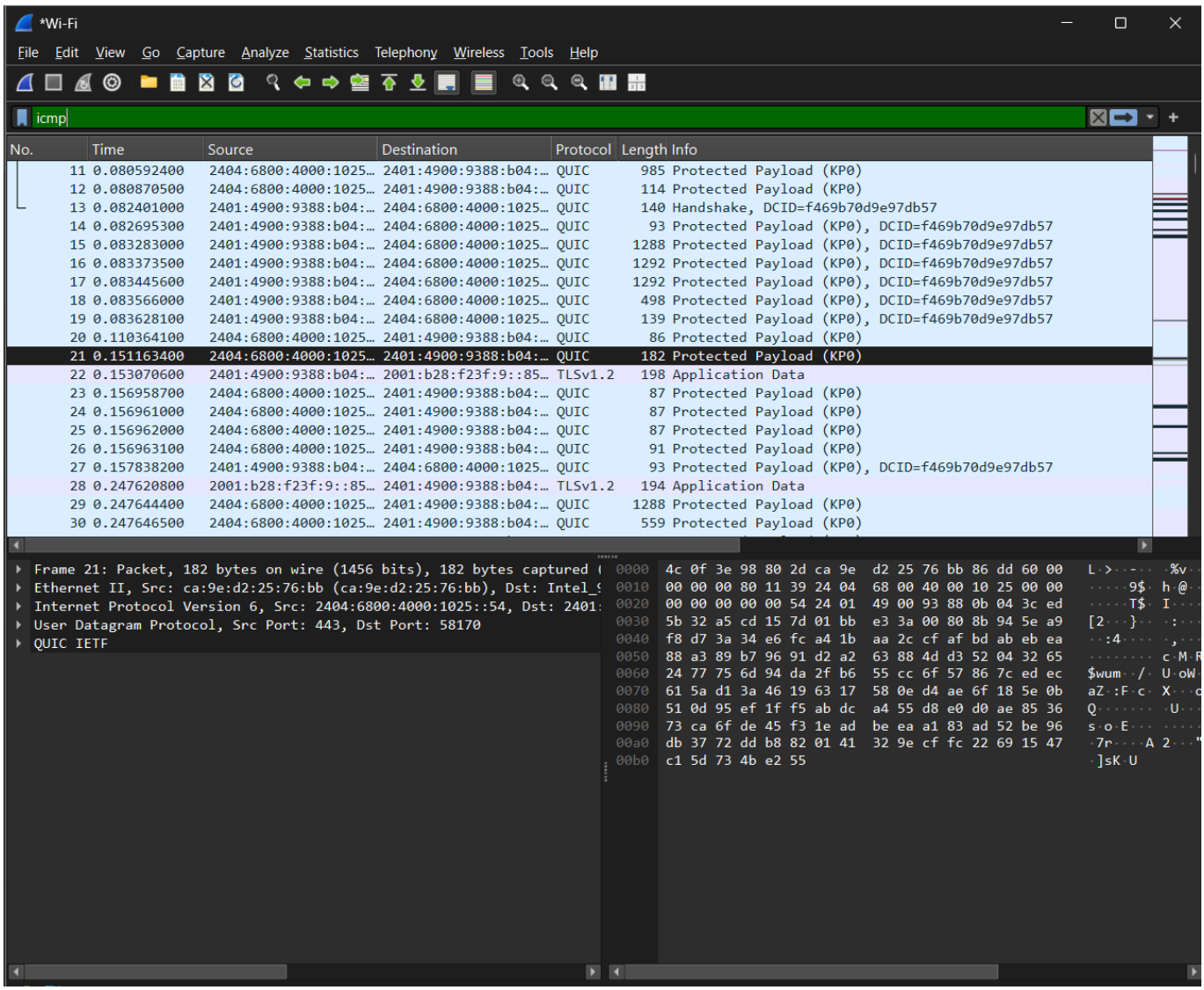
Step 3: Browse a few websites, then stop the capture.



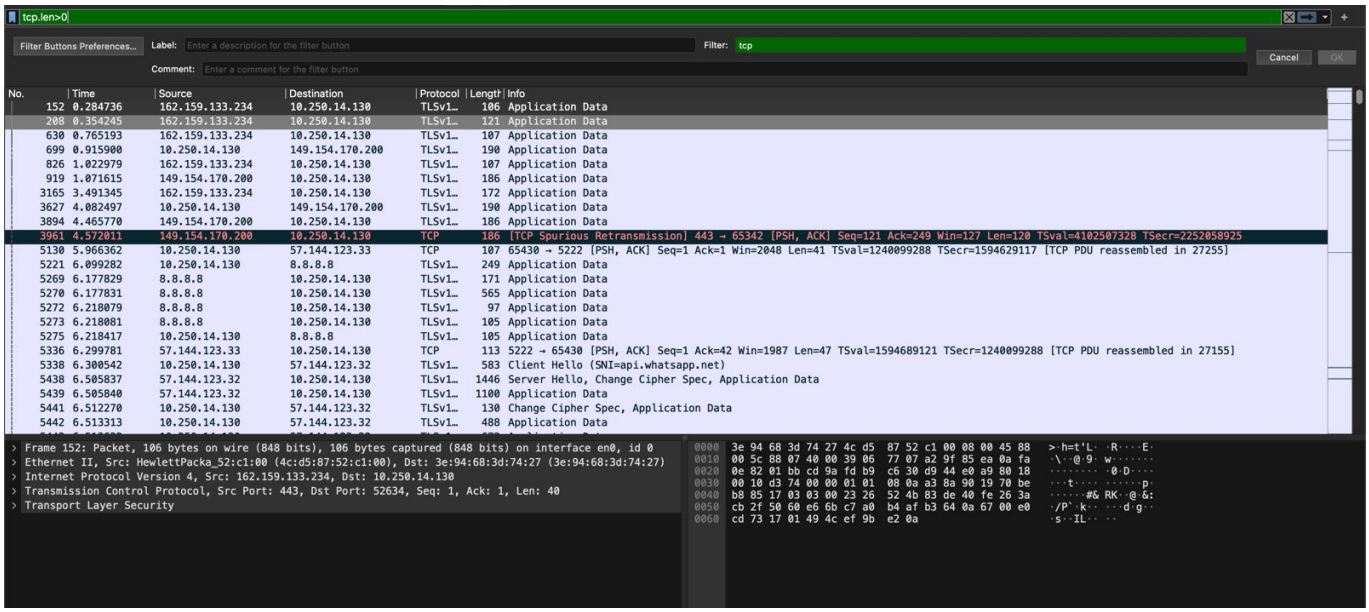
Step 4: Apply the display filters tcp, udp and icmp.




udp
Filter Buttons Preferences...
Label: Enter a description for the filter button
Filter: tcp
Comment: Enter a comment for the filter button
Cancel
OK
No. Time Source Destination Protocol Length Info
2 0.013081 10.250.16.29 224.0.0.251 MDNS 470 Standard query response 0x0000 PTR Ashreeth's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 491…
6 0.014223 fe80::1094:3fae:c1… ff02::fb MDNS 435 Standard query response 0x0000 PTR uditha's MacBook._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 54965 udi…
9 0.014226 fe80::fd6:e9ad:f01… ff02::fb MDNS 337 Standard query response 0x0000 PTR Spotify Desktop Launcher(6)._spotify-connect._tcp.local SRV 0 0 51738 sraagvi.local TXT A 10.25…
10 0.014964 10.250.12.62 224.0.0.251 MDNS 954 Standard query response 0x0000 TXT, cache flush PTR _airplay._tcp.local PTR Nikhil's MacBook Air._airplay._tcp.local TXT, cache fl…
11 0.015735 fe80::dc1b:57a6:7d… ff02::fb MDNS 583 Standard query response 0x0000 PTR, cache flush DESKTOP-VVNEN40._dosvc._tcp.local SRV, cache flush 0 0 7680 DESKTOP-VVNEN40.local …
12 0.015736 10.250.29.11 224.0.0.251 MDNS 499 Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-VVNEN40.local TXT, cache flush A, cache flush 10.250.29.11 AAAA, …
16 0.016387 10.250.22.47 224.0.0.251 MDNS 415 Standard query response 0x0000 PTR MAC's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 49163 MA…
17 0.016682 fe80::4b5:a222:5b9… ff02::fb MDNS 435 Standard query response 0x0000 PTR MAC's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 49163 MA…
23 0.019319 10.250.15.4 224.0.0.251 MDNS 150 Standard query 0x0000 AAAA DS-2CD3056G2-IS20230306AAWRL37490970.local, "QM" question A DS-2CD3056G2-IS20230306AAWRL37490970.local,…
26 0.019589 fe80::ede8:e7:d94d… ff02::fb MDNS 264 Standard query 0x0002 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googlecast._t…
31 0.021530 10.250.26.224 224.0.0.251 MDNS 128 Standard query 0x0000 SRV brother dcp-l2520d series @ g.rohith's macbook air._ipp._tcp.local, "QM" question
32 0.021532 10.250.26.224 224.0.0.251 MDNS 129 Standard query 0x0000 TXT brother dcp-l2520d series @ g.rohith's macbook air._ipps._tcp.local, "QM" question
33 0.022081 10.250.26.224 224.0.0.251 MDNS 129 Standard query 0x0000 SRV brother dcp-l2520d series @ g.rohith's macbook air._ipps._tcp.local, "QM" question
34 0.022082 fe80::e64e:5c50:4f… ff02::fb MDNS 149 Standard query 0x0000 SRV brother dcp-l2520d series @ g.rohith's macbook air._ipps._tcp.local, "QM" question
35 0.022084 10.250.18.223 224.0.0.251 MDNS 440 Standard query response 0x0000 TXT PTR Abhilash's MacBook Air._companion-link._tcp.local TXT, cache flush SRV, cache flush 0 0 617…
36 0.022384 fe80::1872:9bd8:1b… ff02::fb MDNS 460 Standard query response 0x0000 TXT PTR Abhilash's MacBook Air._companion-link._tcp.local TXT, cache flush SRV, cache flush 0 0 617…
37 0.022385 10.250.26.224 224.0.0.251 MDNS 128 Standard query 0x0000 TXT brother dcp-l2520d series @ g.rohith's macbook air._ipp._tcp.local, "QM" question
38 0.023310 fe80::e64e:5c50:4f… ff02::fb MDNS 148 Standard query 0x0000 TXT brother dcp-l2520d series @ g.rohith's macbook air._ipp._tcp.local, "QM" question
39 0.023311 10.250.26.224 224.0.0.251 MDNS 128 Standard query 0x0000 SRV brother dcp-l2520d series @ g.rohith's macbook air._ipp._tcp.local, "QM" question
40 0.023365 fe80::e64e:5c50:4f… ff02::fb MDNS 148 Standard query 0x0000 SRV brother dcp-l2520d series @ g.rohith's macbook air._ipp._tcp.local, "QM" question
42 0.023561 10.250.17.113 224.0.0.251 MDNS 150 Standard query 0x0000 A DS-2CD3056G2-IS20230306AAWRL37490970.local, "QM" question AAAA DS-2CD3056G2-IS20230306AAWRL37490970.local,…
45 0.024287 10.250.26.224 224.0.0.251 MDNS 128 Standard query 0x0000 TXT brother dcp-l2520d series @ g.rohith's macbook air._ipp._tcp.local, "QM" question
46 0.024288 fe80::e64e:5c50:4f… ff02::fb MDNS 148 Standard query 0x0000 TXT brother dcp-l2520d series @ g.rohith's macbook air._ipp._tcp.local, "QM" question
Frame 2: Packet, 470 bytes on wire (3760 bits), 470 bytes captured (3760 bits) on interface en0, id 0
Ethernet II, Src: b2:de:a9:25:f3:08 (b2:de:a9:25:f3:08), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
Internet Protocol Version 4, Src: 10.250.16.29, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Length Info
11 0.080592400 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 985 Protected Payload (KP0)
12 0.080870500 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 114 Protected Payload (KP0)
13 0.082401000 2401:4900:9388:b04:… 2404:6800:4000:1025… QUIC 140 Handshake, DCID=f469b70d9e97db57
14 0.082695300 2401:4900:9388:b04:… 2404:6800:4000:1025… QUIC 93 Protected Payload (KP0), DCID=f469b70d9e97db57
15 0.083283000 2401:4900:9388:b04:… 2404:6800:4000:1025… QUIC 1288 Protected Payload (KP0), DCID=f469b70d9e97db57
16 0.083373500 2401:4900:9388:b04:… 2404:6800:4000:1025… QUIC 1292 Protected Payload (KP0), DCID=f469b70d9e97db57
17 0.083445600 2401:4900:9388:b04:… 2404:6800:4000:1025… QUIC 1292 Protected Payload (KP0), DCID=f469b70d9e97db57
18 0.083566000 2401:4900:9388:b04:… 2404:6800:4000:1025… QUIC 498 Protected Payload (KP0), DCID=f469b70d9e97db57
19 0.083628100 2401:4900:9388:b04:… 2404:6800:4000:1025… QUIC 139 Protected Payload (KP0), DCID=f469b70d9e97db57
20 0.110364100 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 86 Protected Payload (KP0)
21 0.151163400 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 182 Protected Payload (KP0)
22 0.153070600 2401:4900:9388:b04:… 2001:b28:f23f:9::85… TLSv1.2 198 Application Data
23 0.156958700 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 87 Protected Payload (KP0)
24 0.156961000 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 87 Protected Payload (KP0)
25 0.156962000 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 87 Protected Payload (KP0)
26 0.156963100 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 91 Protected Payload (KP0)
27 0.157838200 2401:4900:9388:b04:… 2404:6800:4000:1025… QUIC 93 Protected Payload (KP0), DCID=f469b70d9e97db57
28 0.247620800 2001:b28:f23f:9::85… 2401:4900:9388:b04:… TLSv1.2 194 Application Data
29 0.247644400 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 1288 Protected Payload (KP0)
30 0.247646500 2404:6800:4000:1025… 2401:4900:9388:b04:… QUIC 559 Protected Payload (KP0)
Frame 21: Packet, 182 bytes on wire (1456 bits), 182 bytes captured
Ethernet II, Src: ca:9e:d2:25:76:bb (ca:9e:d2:25:76:bb), Dst: Intel_
Internet Protocol Version 6, Src: 2404:6800:4000:1025::54, Dst: 2401
User Datagram Protocol, Src Port: 443, Dst Port: 58170
QUIC IETF

Step 5: Select a TCP packet and expand Ethernet II, Internet Protocol and TCP.



Step 7: Observation table

Parameter	TCP Packet	UDP Packet
Frame Length	106 bytes	470 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	32 bytes	8 bytes
IP Total Length	92 bytes	456 bytes
Transport Segment Length	72 bytes	436 bytes
Payload Length	40 bytes	428 bytes